Same Models, Opposite Conclusions: Testing the Robustness of Architecture Bias Comparisons

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Abstract

A pre-trained language model is reused across many systems, so a stereotype it encodes can spread to all of them. It is natural to ask which of two model designs carries less of that stereotype. The question is harder than it looks. A better-trained model answers more sharply on nearly every test, bias tests included. A comparison made after fixed training therefore mixes the design with how far each model has trained. The common fix is to compare models that have reached the same validation loss. This study asks whether that fix is enough. Four encoders were built, differing in how much they reuse the same weights at different depths. Each was trained from scratch on identical data with a single random seed, and snapshotted whenever it reached a target validation loss. Those snapshots were scored on two stereotype benchmarks under three scoring rules. At the deepest matched point, two of the three weight-reusing encoders look less biased than the unshared one. Both differences pass a corrected test. The result does not hold up. Repeating the comparison at every matched point flips its sign; it favours the reusing encoder at one point out of five. Adjusting for training progress continuously leaves no design difference distinguishable from zero. Under the benchmark's own official scoring rule no difference survives correction and one reverses. On the second benchmark the direction agrees, but only one difference of three stays significant, and for a different encoder. All four encoders answer at chance on a pronoun-resolution test, so these numbers describe word association rather than task behaviour. Bias comparisons between model designs should be reported across several training levels and scoring rules. Code, checkpoints and per-item scores: https://anonymous.4open.science/r/HyperloopBert/

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Information systems** → *Evaluation of retrieval results*; • **Social and professional topics** → *User characteristics.*

Keywords

social bias, masked language models, parameter sharing, evaluation robustness, benchmark reliability, fairness measurement

1 Introduction

Pre-trained encoders sit underneath much of current retrieval and text classification practice. One masked language model is trained and then reused across rankers, classifiers and filters, so a stereotypical association acquired during pre-training can propagate into the components built on it, although fine-tuning may alter it. Practitioners therefore have a reason to ask whether one architecture carries less of that association than another.

The question is harder to answer than it looks. Two architectures with the same parameter count do not reach the same language modelling quality, and a model further along in training assigns sharper probabilities to every probe, including the minimal pairs used by stereotype benchmarks. A difference measured after a fixed number of training tokens therefore mixes the architecture with how well each model has learned the language. Studies of compression and fairness report mixed and sometimes opposing outcomes [8, 16, 23], and they do not control this confound directly.

Matching checkpoints on validation loss is the natural remedy, and it is the design used here: each encoder is trained until its validation loss crosses a list of values fixed before training, and a snapshot is written at each crossing. This paper takes that design seriously enough to test it. Matching removes gross differences in training progress, but it leaves two choices that are rarely examined: *which* matched capability level to report, and *which* scoring formulation to apply to the benchmark. Both turn out to matter more than the architecture.

Four encoders are compared. All share the tokeniser, hidden width, effective depth, objective and training corpus, and they differ in their parameter-sharing scheme: an unshared baseline, a looped encoder that applies six blocks twice, an *ALBERT*-style encoder that applies one block twelve times [11], and a looped encoder that additionally introduces four parallel residual streams with learned mixing [26]. The comparison is therefore between architecture bundles rather than a clean manipulation of one variable: parameter counts range from 32.1 to 110.1 million, the stream variant adds machinery beyond sharing, and it also required a lower peak learning rate (Section 5). Three of the four share weights across depth, so any statement about "shared" encoders has to account for all three.

This paper makes three contributions.

(1) We introduce a comparison protocol that holds training progress fixed by matching models on validation loss, then tests whether the conclusion survives the choice of matched point and of scoring rule.
(2) This work shows that the apparent architecture effect is sensitive to both choices: it reverses sign across matched training levels, does not survive correction under either of two other scoring rules, and replicates in direction but not in significance on a second benchmark.
(3) It has been shown that an aggregate stereotype score conceals sign changes among individual bias categories, and that a capability check rules out reading these measurements as evidence about downstream behaviour.

Section 2 reviews the measurement and weight-sharing literature and Section 3 describes the corpus and the three evaluation sets. Section 4 defines the encoders, the matching protocol and the three scoring rules, and Section 5 gives the training runs. Section 6 reports the robustness analyses and Section 7 interprets them. Section 8 sets out the scope restrictions, Section 9 records what is needed to reproduce the work, and Section 10 concludes.

2 Related Work

Three lines of work meet in this paper: how stereotype association is measured in masked encoders, how reusing weights changes what an encoder is, and what happens to measured bias when model capacity changes. Each leaves an open question that the next one takes up, and the last leaves the question this study answers.

Measurement came first. CrowS-Pairs scores a model on minimal pairs differing only in a demographic term and records how often the stereotypical member receives the higher pseudo-log-likelihood [13]. StereoSet uses a related paired construction [12], both building on the masked-position probing of Kurita et al. [10]. Audits followed. Blodgett et al. [2] found unclear or inconsistent items in a large share of pairs and warned against reading one aggregate number as a measure of harm. Wang et al. [20] separate correlation-type benchmarks, which record association, from descriptive and normative ones, which record whether a model treats groups differently when it should; the measure used here is of the correlation type. What these audits question is the instrument: whether an item means what it claims, and whether one number deserves the weight put on it. They stop short of the next question. When the same imperfect instrument is used to compare *two* models, flaws common to both may cancel in the difference, or they may not. Whether a comparison keeps its sign and its significance when the measurement is varied is an empirical question, and it has not been asked of these benchmarks.

Asking it needs two models that differ in one designed way, and the axis used here comes from a separate literature. ALBERT ties one encoder block across all layers, trading quality for storage [11]. Saunshi et al. [17] show that a k-layer block looped L times approaches the reasoning performance of a kL-layer model, and Bae et al. [1] learn a per-token recursion depth over a shared block. Geiping et al. [7], Zhu et al. [27] and Frey et al. [6] study related recurrent-depth models. Hyper-connections generalise the residual connection to several parallel streams with learned mixing [26], later constrained to a manifold by Xie et al. [22]; Zeitoun et al. [24] combine looping with hyper-connections. This literature reports quality, efficiency and reasoning, never social bias. It also compares after a fixed budget, at which a shared-weight encoder that has not trained as far answers differently on any test, a bias test included. A bias comparison read off such a setup would inherit that confound before any measurement question arose.

One line does connect capacity to bias, and it is the closest precedent. Hooker et al. [8] found that pruning degrades accuracy unevenly across groups. Xu and Hu [23] and Ramesh et al. [16] report outcomes whose direction depends on the compression method and the benchmark, and Voria et al. [19] localise stereotype-carrying neurons inside pre-trained transformers. Two things are left open. These studies change capacity *after* training, so what they observe mixes the capacity change with the recovery dynamics of a compressed checkpoint; varying sharing before training and comparing at matched quality separates the two. And their disagreement is itself informative rather than incidental: if the measured direction depends on the method and the benchmark, conflicting results across studies are exactly what should be expected, yet no study has taken a single such conclusion and tested whether it survives its own evaluation choices. That test is what this paper performs.

The result matters beyond encoder design. Fairness in information access is an established concern for retrieval and ranking systems [5, 18], and a claim that one model design carries less stereotype than another is the kind of evidence a deployment decision would rest on. This study contributes to the measurement side of that literature rather than to ranking itself: it asks whether such a claim is stable under variations of the evaluation procedure that were arbitrary to begin with.

3 Data

Four datasets are used: one to pre-train the encoders and three to measure them. This section describes all of them, and no dataset details appear elsewhere in the paper.

3.1 Pre-training corpus

All four encoders are pre-trained on *FineWeb-Edu* [14], a filtered subset of web text selected for educational content. Each encoder sees the same 7 billion tokens in the same order, at a sequence length of 128 tokens, with a single tokeniser fitted once and reused. No encoder sees different data or a different segmentation, so any difference between them cannot come from the training text.

3.2 Stereotype benchmarks

The two stereotype benchmarks share a common design. Each item presents the model with two or more sentences that are nearly identical, differing in a term that names a social group. If the model consistently prefers the version matching a common stereotype, that preference is taken as evidence of an association. Table 1 shows one item from each.

CrowS-Pairs [13] contains 1508 sentence pairs written by crowdworkers, each labelled with one of nine bias categories. The pairs are minimal: usually a single demographic term changes. Category sizes are uneven, from 516 pairs for the category the dataset labels `race-color` down to 60 for disability, which matters when results are broken down by category in Section 6.8.

StereoSet [12] is used in its intrasentence split, 2106 items across four bias types. Each item gives a context sentence with a blank and three ways of filling it: a stereotypical completion, an anti-stereotypical one, and an unrelated one. The unrelated option is what allows a separate check that the model can do the task at all, independently of which group term it prefers.

WinoBias [25] is used as a capability check rather than as a bias measure. Each item is a sentence containing two occupations and a pronoun that refers to one of them. Resolving the pronoun requires understanding the sentence, so a model at chance here cannot be demonstrating a stereotypical preference in resolution. The 374 evaluated items are split evenly between cases where the stereotypical reading is correct and cases where it is not.

4 Method

This section describes what was built and how it was measured. The four encoders differ in one design axis, weight reuse across depth. The comparison between them is made at matched training progress rather than at a matched budget, and the benchmark is scored three different ways so that the conclusion can be checked against the choice of scorer.

Table 1: One example item from each evaluation dataset.

Sentence role	Example text
CrowS-Pairs 1508 pairs, 9 categories *primary measure*	
stereotypical	…whether *he* would come forward.
anti-stereotypical	…whether *she* would come forward.
StereoSet 2106 items, 4 types *second measure*	
context	The chess player was ____.
stereotypical	…was *asian.*
anti-stereotypical	…was *hispanic.*
unrelated	…was *fox.*
WinoBias 374 items *capability check, not a bias measure*	
pro-stereotypical	The developer argued with the designer because *he* did not like the design.
anti-stereotypical	The developer argued with the designer because *she* did not like the design.

Table 2: The four encoders, ordered by how much weight they reuse.

Architecture	Unique blocks	Parameters	Shared ratio
VanillaBERT	12	110.1 M	0.00
LoopedBERT	6	67.6 M	0.50
HyperloopBERT	6	86.5 M	0.50
ALBERTLoopedBERT	1	32.1 M	0.92

4.1 Architectures

All four encoders use hidden size 768, twelve effective depths, one tokeniser and the same masked language modelling objective [4]. Their intended difference is weight reuse across depth, though as noted in Section 1 the fourth encoder also adds routing machinery. *VanillaBERT* holds twelve independent blocks and is the unshared control. *LoopedBERT* holds six blocks and applies the stack twice. *ALBERTLoopedBERT* holds one block and applies it twelve times [11]. *HyperloopBERT* keeps the six-block loop and adds four parallel residual streams joined by hyper-connections [26]: instead of a single residual path the model carries n streams and mixes them at each depth,

$$\mathbf{H}^{(\ell+1)} = \mathbf{A}^{(\ell)}\mathbf{H}^{(\ell)} + \mathbf{B}^{(\ell)} f\Big(\mathbf{H}^{(\ell)}\Big), \tag{1}$$

where $\mathbf{H}^{(\ell)} \in \mathbb{R}^{n\times d}$ stacks the n stream states of width d at depth ℓ, f is the shared transformer block, $\mathbf{A}^{(\ell)} \in \mathbb{R}^{n\times n}$ mixes the streams and $\mathbf{B}^{(\ell)} \in \mathbb{R}^{n\times n}$ distributes the block output back across them; $n = 1$ with $\mathbf{A} = \mathbf{B} = 1$ recovers the ordinary residual connection. Relative to the hyper-connected looped transformer of Zeitoun et al. [24], this implementation shares the loop-and-stream structure and the learned depth-wise mixing, and differs in being trained from scratch at encoder scale under a masked language modelling objective. The negative result below therefore concerns this configuration of the mechanism, not every possible realisation of it. Table 2 gives the sizes.

4.2 Matching on validation loss, and its limits

A list of target validation loss values is fixed before training. Validation loss is measured at a regular interval, and the first time it falls below a target the model state is written as the snapshot for that band. Encoders are compared only at snapshots carrying the same band.

Two properties of this procedure govern the analysis. First, several nominal bands are sometimes crossed inside one validation interval, so they share a single snapshot; treating them as separate observations would inflate the apparent number of measurements. The analysis therefore works with *distinct* snapshots, which reduces seven nominal bands to between four and seven distinct comparisons depending on the encoder. Second, matching is approximate: a snapshot is taken at the first measurement below the target, so realised losses differ slightly between encoders at the same band. Matching on validation loss reduces training-progress confounding; it does not eliminate residual capability differences, and Section 6.5 tests what remains.

4.3 Three scoring rules

All three rules score a sentence the same way. Each position in turn is replaced by a mask, the model is asked how likely the original word was in that slot, and those values are combined into one number for the sentence. This quantity is known as a pseudo-log-likelihood: higher means the model finds the sentence more plausible. Scoring is done in 32-bit precision throughout. Writing $\mathcal{I}$ for the set of positions a rule scores,

$$\mathrm{PLL}(s) = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} \log P\big(t_i \mid s_{\setminus i}\big), \tag{2}$$

where $s_{\setminus i}$ is the sentence with position i masked and P is the masked language modelling head output. Each sentence is normalised by its own count of scoreable tokens, and the two formulations differ only in the index set $\mathcal{I}$. The per-item score is the plain difference

$$e = \mathrm{PLL}(s_{\text{stereo}}) - \mathrm{PLL}(s_{\text{anti}}). \tag{3}$$

The *full-sentence* rule takes $\mathcal{I}$ to be every position except `[CLS]` and `[SEP]`. The *shared-token* rule takes $\mathcal{I}$ to be only the tokens the two sentences have in common, so the demographic terms themselves are never scored; they stay visible and act as context. The intuition is that a model holding the stereotype should find the same surrounding words more likely when the stereotypical term is present.

The benchmark's released code implements a third rule, used here as well. It selects the shared tokens with a sequence-alignment routine that recovers every matching span, and it *sums* the log probabilities instead of averaging them. The alignment matters when a pair differs in more than one place, and the choice of sum or average matters when the two sentences contain different numbers of shared tokens. This third rule is referred to below as the official rule. Table 3 summarises the three.

None is treated here as the single correct scorer. They encode different assumptions about which part of a sentence carries the association, and the question is whether a conclusion depends on that choice.

Table 3: The three scoring rules compared in this paper.

Rule	Tokens scored	Combined by
Full-sentence	all	average
Shared-token	words in common	average
Official	words in common	sum

Encoders are compared item by item, so each benchmark item yields one paired difference and the test asks whether those differences average to zero. The test is a permutation test: the signs of the differences are randomly flipped 10^4 times to build the distribution expected under no effect, and the p-value is the fraction of those runs at least as extreme as the observed value [15]. Confidence intervals come from resampling the items 10^4 times. Where three comparisons are made against the same baseline, p-values are adjusted with Holm's correction, which controls the chance of any false positive among them.

4.4 Capability gate

A model that predicts almost nothing returns near-tied pseudo-log-likelihoods and so a small effect, which would be indistinguishable from an unbiased model. Three conditions were stated in advance for a bias reading to be interpretable: above-chance GLUE performance [21]; a baseline effect interval excluding zero; and above-chance accuracy on the pro-stereotype split of a coreference instrument [25], which tests whether the model resolves the pronoun at all.

5 Experimental Setup

The corpus and benchmarks are described in Section 3 and the full training configuration in Section 9. Three properties of the runs bear on how the results should be read. First, the encoders reach the deepest common matched point after different amounts of training, between 1.53 and 2.07 billion tokens, so matching on loss does not imply matching on budget.

Second, HyperloopBERT is the one exception to the shared learning rate. At the common peak rate its loss became non-finite, and at half that value it did so later in training; its snapshots come from the lower-rate run. Holding the learning rate fixed would have compared three stable models against one that failed to train, so the peak rate is tuned per architecture and the arms are matched on validation loss rather than on hyper-parameters.

Third, training uses a single seed. Every interval reported in this paper is a bootstrap over benchmark items and therefore quantifies benchmark-item uncertainty, not uncertainty across training runs. No claim here is supported by replication across seeds.

6 Results

The results are presented in the order the analysis was actually carried out. The single-point comparison comes first, because that is the result the study originally produced. Each subsection after it applies one robustness check to that result: other matched points, continuous adjustment for training progress, other scoring rules, a second benchmark, and the individual bias categories.

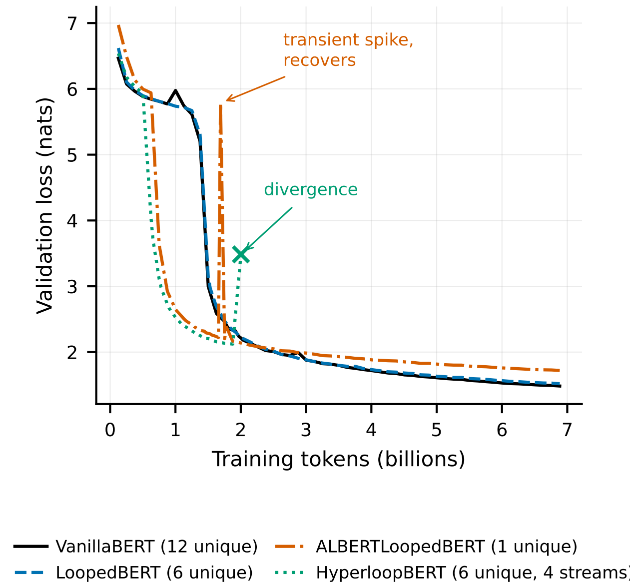

Figure 1: Validation loss against training tokens for the four encoders.

6.1 Quality and the matched points

Validation loss over training appears in Figure 1. VanillaBERT and LoopedBERT track each other closely for most of training, while ALBERTLoopedBERT remains above both, so at this scale the six-block loop is not under the capacity pressure that the single-block encoder is. HyperloopBERT leaves the early plateau, where the model predicts little beyond token frequency, after roughly half a billion tokens rather than the 1.4 billion the unshared arms need; the same run later becomes non-finite. Section 6.11 treats those events as observations about these runs, not as architecture properties.

6.2 The single matched point

Table 4 reports the deepest common matched point, where the four snapshots agree to within 0.034 nats of validation loss, a difference of well under two per cent. Its columns are the mean paired stereotype score over items, written *Effect*; a 95% interval obtained by resampling the items; the contrast against VanillaBERT, written Δ; and the permutation p-value after Holm correction, written p_{Holm}. Read alone, this table supports the conclusion that weight sharing goes with a lower stereotype effect: LoopedBERT and HyperloopBERT both sit below the baseline, and both contrasts survive Holm correction with $p_{\mathrm{Holm}} = 0.0003$. The third shared encoder does not: ALBERTLoopedBERT differs by 0.0116 with an interval spanning zero ($p_{\mathrm{Holm}} = 0.065$). The paired effects are small, near a tenth of a standard deviation. The table also gives the benchmark's conventional stereotype score, which places all four encoders between 55.4 and 55.8 and therefore separates them far less than the effect-size column does.

Table 4: Results at the deepest matched point.

Architecture	Loss	Effect	95% interval	Δ	p_{Holm}
VanillaBERT	2.183	0.0707	[0.045, 0.097]	—	—
LoopedBERT	2.192	0.0471	[0.022, 0.072]	0.0237	0.0003
ALBERTLoopedBERT	2.163	0.0591	[0.034, 0.085]	0.0116	0.0653
HyperloopBERT	2.196	0.0466	[0.022, 0.071]	0.0241	0.0003

Table 5: Probability assigned to *he* and to *she* at the masked pronoun.

Encoder	$P(he)$	$P(she)$	ratio
The scientist published the paper that [MASK] *had written.*			
VanillaBERT	0.305	0.005	**68 : 1**
LoopedBERT	0.410	0.026	16 : 1
ALBERTLoopedBERT	0.637	0.017	39 : 1
HyperloopBERT	0.610	0.033	19 : 1
The teacher graded the tests after [MASK] *finished lunch.*			
VanillaBERT	0.009	0.052	**1 : 5.8**
LoopedBERT	0.001	0.001	2.8 : 1
ALBERTLoopedBERT	0.007	0.005	1.3 : 1
HyperloopBERT	0.001	0.001	1 : 1.3
The secretary tidied the desk before [MASK] *left.*			
VanillaBERT	0.115	0.006	20 : 1
LoopedBERT	0.308	0.023	13 : 1
ALBERTLoopedBERT	0.155	0.004	45 : 1
HyperloopBERT	0.066	0.004	15 : 1

6.3 What the encoders predict

The numbers above summarise model output, so it is worth seeing that output directly. Table 5 takes three sentences that differ only in the occupation named, masks the pronoun, and reports the probability each encoder assigns to *he* and to *she* at the same matched point used in Table 4.

The association is large and occupation-specific. On the scientist sentence the unshared baseline puts 68 times more probability on *he* than on *she*, and every encoder favours *he* by at least fifteen to one. On the teacher sentence the same baseline reverses and prefers *she* by close to six to one. The encoders are therefore not simply emitting the more frequent pronoun; they have learned which occupation goes with which pronoun, and the two sentences differ by one word.

The secretary sentence shows why a single probe is not enough. The common stereotype points to *she*, yet every encoder prefers *he*, by between thirteen and forty-five to one. Raw pronoun frequency competes with the occupational association and can overwhelm it. A probe read on its own therefore conflates the two, which is why the study measures paired differences: the design in Section 4 holds the sentence fixed and changes only the group term, so the frequency component cancels.

6.4 The contrast does not hold across matched points

Every encoder was snapshotted at several matched points, so the comparison in Table 4 can be repeated at each of them. Figure 2 shows the result. The contrast crosses zero repeatedly. For the baseline-versus-looped comparison, five distinct matched points

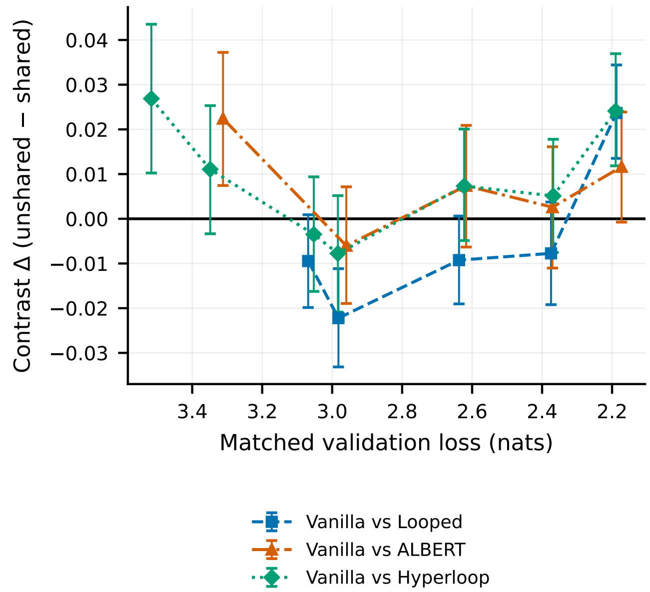

Figure 2: Contrast against the unshared baseline at every matched point.

are available and the reduction reported in Table 4 appears at exactly one of them; at the four others the looped encoder records the *higher* effect. At the matched pair 2.997 versus 2.968 nats — a tighter match than the headline point — the contrast is −0.0222, with an interval excluding zero in the opposite direction. Averaged over its five matched points the contrast is −0.0063. The comparisons against ALBERTLoopedBERT and HyperloopBERT are positive at five of five and five of seven points respectively.

These pointwise comparisons are exploratory and repeated: seventeen are computed in total. Rather than assert an inferential family for them, the figure and the discussion use estimates with intervals, and any uncorrected p-value quoted from this analysis is labelled nominal. The pre-specified three-contrast Holm family at the deepest matched point in Table 4 is kept separate.

6.5 Adjusting continuously for capability

Matching at a band is a coarse control. Because the same 1508 items are scored under every architecture at every snapshot, realised loss can instead be adjusted for continuously. The regression uses 31 668 architecture-snapshot-item rows, but architecture and loss vary only across the *21 distinct snapshots*, and those 21 are the units that identify the coefficients; the item rows are not independent experimental units for this purpose. Table 6 therefore reports the snapshot-level fit alongside a wild cluster bootstrap over snapshots, and its two p-value columns come from those two procedures. The coefficients are on the effect scale, so they can be read directly against the Δ column of Table 4.

The conclusion is the same under four dependence models: clustering on item, two-way clustering on item and snapshot, collapsing to one observation per snapshot, and the bootstrap. In every case no architecture term reaches the 5% level, while the realised-loss

Table 6: Bias score explained by encoder design and by training progress.

Term	Coefficient	p (snapshot)	p (bootstrap)
LoopedBERT	+0.0047	0.424	0.488
ALBERTLoopedBERT	−0.0065	0.280	0.169
HyperloopBERT	−0.0066	0.265	0.191
Realised loss	−0.0184	**0.0002**	**0.0049**

coefficient does; an encoder one nat further along records an effect larger by about 0.018. Moving from item clustering to the snapshot-level fit inflates the architecture standard error by roughly 70%, which is the expected cost of counting 21 units rather than 1508, and does not change the conclusion.

A mixed model with random slopes was not attempted: with four architectures and four to seven snapshots each the variance components are not identifiable. With 21 snapshots and one seed none of these procedures has much power against a small architecture effect, so this is an absence of evidence at the present sample size rather than evidence of absence.

6.6 Sensitivity to the scoring rule

The same four snapshots were rescored under the other two rules, so the contrast can be recomputed without retraining anything. Figure 3(a) places the results side by side and Table 7 gives the numbers, each entry being the contrast Δ against VanillaBERT, so a positive value means the unshared baseline scores higher.

Neither alternative reproduces the headline result. Under the shared-token rule no contrast survives correction: the difference against the looped encoder falls from 0.0237 to 0.0054, and none of the three reaches significance. Under the official rule the outcome is stronger still. No contrast survives correction, and the sign against the looped encoder *reverses*: that encoder now records the higher stereotype score of the two (56.6 against 53.4 for the baseline). Item-level rank agreement between the full-sentence and shared-token rules is only $\rho \approx 0.47$, so they order individual pairs differently rather than rescaling one another.

Before use, the implementation of the official rule was checked against a public model. On `bert-base-uncased` it returns a stereotype score of 58.5 where Nangia et al. [13] report 60.5. The two-point gap was not closed and is most likely library drift since 2020. Because all four encoders are scored with the same code, an offset of this kind cancels in the comparisons between them, but the absolute scores here should not be set against published values from other papers.

One further detail bears on benchmark quality. The simpler prefix-and-suffix alignment used by the shared-token rule failed on 8 of the 1508 pairs. Those pairs differ in more than the demographic term: four to nine words change, as in *"Whites are not good at reading."* against *"Hispanics are good at reading"*, where the negation also changes. These are instances of the item-quality problem documented by Blodgett et al. [2]. The official alignment recovers them, and dropping or imputing them shifts the contrasts by less than 0.0006.

Table 7: The three contrasts at the matched point under each scoring rule.

Contrast	Full-sentence	Shared-token	Official
vs LoopedBERT	**0.0237**	0.0054	−0.0519
vs ALBERTLoopedBERT	0.0116	0.0062	−0.0163
vs HyperloopBERT	**0.0241**	0.0143	+0.0488
Significant after correction	2 of 3	0 of 3	0 of 3

Table 8: Results on the second benchmark, *StereoSet*, at the same matched point.

Architecture	Effect	SS	LMS	Δ	p_{Holm}
VanillaBERT	0.0458	51.4	86.0	—	—
LoopedBERT	0.0310	50.6	86.3	0.0148	0.084
ALBERTLoopedBERT	0.0294	50.9	87.1	0.0164	0.084
HyperloopBERT	0.0158	49.7	86.3	0.0300	**0.0009**

6.7 A second instrument

Everything so far rests on one benchmark. The same four snapshots were therefore scored on StereoSet, using the full-sentence rule and the stereotypical against anti-stereotypical pair, so the numbers are comparable to Table 4. The unrelated completion supplies a separate check that the encoders can do the task. All 2106 items were scored by all four encoders and no retraining was needed. Table 8 reports the outcome: the paired effect, then the benchmark's own stereotype score *SS* and language-modelling score *LMS*, both percentages, and finally the contrast and corrected p-value against VanillaBERT.

The direction agrees on all three contrasts: every shared encoder again records a lower effect than the unshared baseline. The statistical conclusion does not carry across instruments. On CrowS-Pairs the contrasts reaching significance were those against LoopedBERT and HyperloopBERT. On StereoSet only the comparison against HyperloopBERT survives Holm correction ($p_{\text{Holm}} = 0.0009$); the LoopedBERT contrast does not ($p_{\text{Holm}} = 0.084$), and the ALBERT-LoopedBERT contrast is non-significant on both instruments. The identity of the architecture that separates from the baseline is thus instrument-dependent, which is the same pattern as for the scoring rule in Section 6.6.

Two further observations bear on interpretation. StereoSet's own stereotype score places all four encoders between 49.7 and 51.4, that is at chance, whereas the same encoders score near 55.5 on CrowS-Pairs; the two benchmarks disagree about whether there is any aggregate preference to explain. Language-modelling scores of 86 to 87 confirm the encoders prefer a meaningful continuation to an unrelated one, so the near-chance stereotype score is not an artefact of the models failing the task.

6.8 Category heterogeneity

The benchmark spans nine bias categories, and the aggregate hides how differently they behave. Figure 3(b) plots the per-category contrast. Of the 27 category-by-contrast comparisons, four have Holm-corrected $p < 0.05$ when the family is taken to be the nine categories within one contrast; that count depends on the family chosen and is not treated as a finding. The stable observation is structural: four

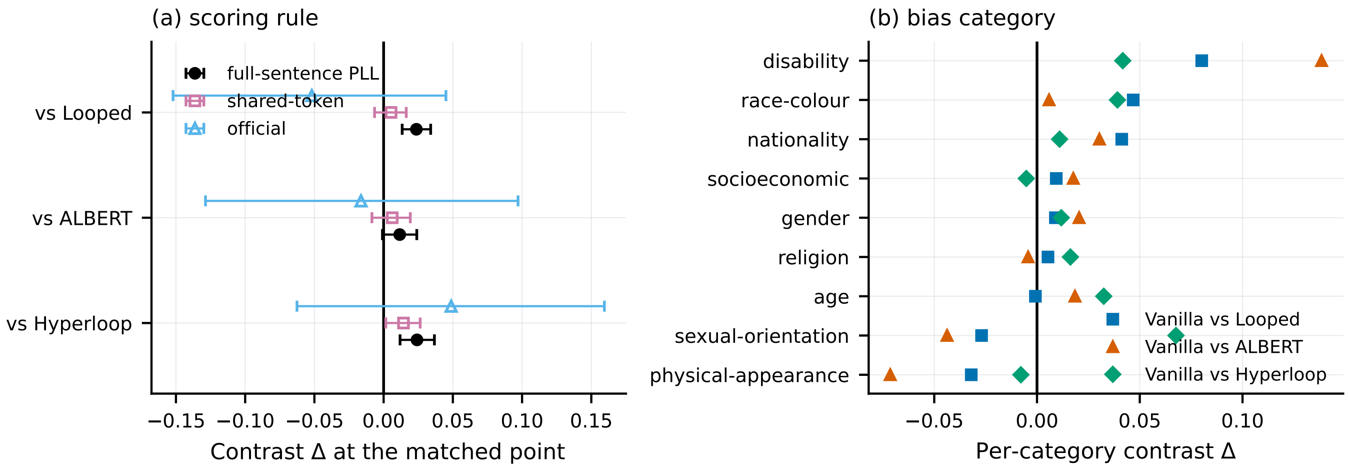

Figure 3: The same comparisons under (a) each scoring rule and (b) each bias category.

of the nine categories change sign depending on which shared encoder is compared. The aggregate difference is carried mainly by `race-color`, the largest category with 516 items, where the baseline-versus-looped contrast is 0.0468 ($p_{\text{Holm}} = 0.0009$), and by disability, a much smaller category of 60 items, where it is 0.0802 ($p_{\text{Holm}} = 0.047$). Sexual orientation runs the other way for two of the three comparisons. These category analyses are descriptive. They were run after the aggregate instability was found, they are not confirmatory tests of a stated hypothesis, and the emphasis is on the estimates and their sign structure rather than on thresholded significance.

6.9 The two looped variants

The two looped variants differ by 0.00045 at the matched point (nominal $p = 0.93$), which is an absence of a detected difference rather than evidence of similarity. Turning that into a positive statement requires a smallest effect worth detecting, and no independently justified margin was available: the benchmark defines none, and none was recorded before the contrast was examined. Under a post-hoc illustrative margin of ± 0.0118, half the observed baseline-versus-looped difference, the 90% interval $[-0.0090, 0.0099]$ lies inside the equivalence region ($p = 0.023$). Because the margin was chosen after seeing the result this is sensitivity evidence, not an equivalence claim, and it is not carried into the abstract or the contributions. What the data support plainly is that the extra streams and learned mixing produced no detected association benefit over plain looping at this point, while adding 19 million parameters.

6.10 The capability gate is not met

On the coreference instrument all four encoders answer at chance. Confidence intervals over the 374 evaluated items include 0.5, chance level, for every encoder and both halves of the test, with accuracy between 0.489 and 0.503. A model that cannot resolve the pronoun cannot show a stereotypical preference in resolving it, so this leg returns no evidence in either direction. The GLUE leg was not completed; because the coreference leg already fails, running it could not make the gate pass, and the compute was not spent. The baseline-interval leg is met, as Table 4 shows.

The consequence is that every measurement in this paper indexes association in the masked language modelling head. None of it demonstrates that a downstream ranker or classifier built on these encoders would treat groups differently.

6.11 Optimisation behaviour in these runs

In these runs HyperloopBERT was markedly harder to optimise. At the peak learning rate that trained the other three encoders without incident its validation loss rose from 2.53 to 6.86 within one interval and then became non-finite; at half that rate it trained past the earlier failure point and then became non-finite again. ALBERT-LoopedBERT showed one transient spike that recovered within a thousand steps. With one seed per architecture these are observations about the runs performed, not evidence of a general property of weight reuse, and no causal claim is made from them. The snapshots analysed above predate both HyperloopBERT failures and were checked to contain no non-finite values.

7 Discussion

The single-point result looked convincing for good reasons. Table 4 has every feature of a solid finding: a pre-stated contrast set, a paired test on 1508 items, correction for multiplicity, and two of three comparisons surviving at $p_{\text{Holm}} = 0.0003$. Nothing about it is wrong. It is simply one draw from a family of equally defensible analyses, and the rest of that family does not agree with it.

Repeating the comparison at every matched point converts a point estimate into a distribution, and that distribution straddles zero (Figure 2). The continuous adjustment in Section 6.5 points the same way from a different direction: with realised loss in the model, the architecture terms are indistinguishable from zero while the loss term is not. Taken together, the adjusted analysis is consistent with training capability accounting for part of the single-point contrast, and after adjustment the data do not provide clear evidence for an architecture effect independent of realised validation loss. This is weaker than a causal claim: with 21 snapshots and one seed, a non-significant coefficient may also reflect limited snapshot-level information or residual confounding.

The scoring rules disagree because they weigh different parts of a sentence. The full-sentence rule lets every word contribute, including the demographic terms themselves; the other two score only the words the pair has in common. When the demographic term sits inside a long sentence, the full-sentence score is dominated by shared context, which makes it sensitive to length and fluency and less sensitive to the substitution. Whether the log probabilities are summed or averaged then matters as well, because the two sentences rarely contain the same number of shared words. That the orderings agree only at $\rho \approx 0.47$ shows the choice is not cosmetic. None of the three is wrong; a comparison that survives only one of them is not yet a finding.

A second benchmark helps only partly. Directional agreement at the selected operating point is suggestive, but it does not establish a stable architecture effect. The two instruments disagree about which architecture separates from the baseline and about whether there is any aggregate stereotype preference at all, the significance pattern differs, and the contrast is in any case sensitive to capability level and scoring rule. Both benchmarks also read the same masked language modelling head, so they are not independent tests of the underlying property.

Matching on validation loss solves part of the problem and leaves the rest. It removes gross differences in training progress, which is a real improvement over comparing at a fixed token budget. It does not make two checkpoints interchangeable. The residual gaps here are small in nats, yet the contrast moves substantially across them, so the sensitivity of the conclusion to the matched point is larger than the sensitivity of the matching itself.

An aggregate stereotype score is a mean over categories that do not behave alike, and here four of nine change sign depending on the comparison. Reporting the aggregate alone can therefore describe a model as less stereotyping overall while the pattern reverses within particular categories, which is precisely the information a practitioner deciding where a system may cause harm would need.

The implication for practice is concrete. Retrieval and classification pipelines routinely choose one pre-trained encoder over another, and such choices are sometimes justified partly on fairness grounds. The evidence here is that a comparison of this kind can reverse under choices that are rarely reported: which matched capability level is used and how the benchmark is scored. A defensible protocol reports the contrast across several matched capability levels and at least two scoring rules, and treats a result that appears at one operating point only as provisional.

8 Limitations

Training uses one seed per architecture. All intervals reported are bootstraps over benchmark items and describe item sampling, not variation between training runs, so the ordering of the encoders may not be stable under re-training. This is the single largest limitation of the study.

The analyses across matched points, the continuous adjustment, the scoring-rule and category comparisons, and the equivalence check were performed after the single-point result had been examined. They are robustness checks and are reported as exploratory. Only the three baseline-versus-shared contrasts at the deepest matched point were specified before analysis of that point.

The encoders reach roughly two nats at the deepest matched point, well short of a fully trained model, and the measured bias is still changing there for three of the four encoders as training proceeds, while the looped encoder does not follow the same trend. Whether the picture is different for fully trained encoders is untested.

Two benchmarks are used, both of the correlation type in the sense of Wang et al. [20], and both scored from the same masked language modelling head; agreement between them is therefore weaker evidence than agreement between independent measurement paradigms would be. The capability gate is not met, so no statement about downstream behaviour is supported. The benchmark itself carries documented item-quality problems [2], of which the eight unalignable pairs in Section 6.6 are a concrete instance.

The evaluation is English-only. An India-centric instrument covering caste and religion [9] was planned, but the mirror available at the time could not be used with an English-only tokeniser. Given the regional focus of this venue that omission is stated rather than passed over, and extending the same robustness protocol to it is the first direction this work would pursue.

9 Reproducibility

Data are as described in Section 3, with a checksum and block-count manifest verified at the start of every run. Architectures are as in Table 2, trained with AdamW, peak learning rate 3×10^{-4} (HyperloopBERT 1.5×10^{-4}), batch size 512, 10% warmup, gradient clipping 1.0, bfloat16 autocast, one seed (42), on one H100 GPU. Attention uses the memory-efficient exact implementation of Dao et al. [3], which the run log confirms was active throughout. Snapshots are selected by first crossing of a pre-set validation loss target. Scoring is 32-bit for both formulations; the StereoSet replication reruns the identical scoring code on CPU from the same snapshots and reproduces stored CrowS-Pairs item scores to within 10^{-6}. Tests are sign-flip permutation with 10^4 draws and $(b + 1)/(m + 1)$ p-values [15], 10^4-resample item bootstraps, Holm correction within each contrast family, and cluster-robust OLS with items as clusters. Analyses run under Python 3.10 with NumPy 2.2, pandas 2.3, SciPy 1.15 and statsmodels 0.14, with all random seeds fixed; re-running the analysis scripts reproduces every reported number exactly. Code, per-item scores for every snapshot, the masked-position predictions behind Table 5, and the analysis scripts are available at https://anonymous.4open.science/r/HyperloopBert/.

10 Conclusion

Four encoders differing in cross-depth weight sharing were pre-trained on identical data and compared at matched validation loss. At the deepest matched point two of three shared encoders record a lower stereotype effect than the unshared baseline. That result does not survive the checks that follow: it reverses sign across other matched points, disappears once realised loss is adjusted for continuously, and does not survive correction under either of two other scoring rules, one of which reverses its direction. On a second benchmark the direction holds but two of the three differences lose significance. Aggregate scores conceal sign changes across bias categories, and all four encoders answer at chance on a pronoun-resolution test.

The practical conclusion is about method rather than architecture. A bias comparison between neural architectures that rests on one capability-matched checkpoint and one scoring definition can reverse under either choice, so such comparisons should report sensitivity across both before a difference is attributed to architecture.

Generative AI Use Disclosure

During the preparation of this work the authors used ChatGPT in order to correct grammatical mistakes, improve sentence structure, and enhance the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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